Matthew DeCicco (TS-SCI Eligible)

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Education

Florida Polytechnic University — Lakeland, FL — May 2024

Bachelor of Science in Mechanical Engineering (Aerospace), ABET accredited program — GPA: 3.87/4.0

Skills

Software:	SolidWorks (CSWP Certified) • Onshape • Mission Planner • MATLAB • ANSYS • FlightStream CFD• XFLOW5
Programming:	C/C++ • Python • Lua
Technical:	ArduPlane • Arduino • Raspberry Pi • CAM/CNC • DAQ • MANET

Experience

Aeromechanical Engineer 2 — AEVEX Aerospace, Tampa, FL — December 2025 — Present

- Serving as acting lead engineer, directing a team of 5 engineers on a Group 3 ISR platform.
- Performed aerodynamic analysis and specified wing geometry, meeting objective stall, cruise performance, and range requirements.
- Supported mechanical design of the new composite wing, including spar and section layout. Worked closely with the vendor to create molds and first articles. Oversaw the testing of the first articles and led the subsequent optimizations.
- Incorporated flight operations & FSR feedback into development to improve usability, including adding a header tank to the fuel system, integrating shore power, and creating operational calculators for parachute recovery and launch fuel load.
- Leading a 5-flight incremental test campaign, coordinating Systems and Flight Test Engineers to track progress against platform requirements, reduce risk, and capture engineering data used to improve system design.
- Partnered with AEVEX's navigation and autonomy branch to develop a custom software stack, integrating an alternate PNT solution (enhanced dead reckoning) with Trillium camera interfacing to deliver GPS-denied capability.

Aeromechanical Engineer 1 — AEVEX Aerospace, Tampa, FL — August 2024 — December 2025

- Integrated MANET networking, alternative PNT, and RATO launch capability onto a Group 3 delta-wing UAS, expanding operational flexibility in GPS-denied environments.
- Designed and built an automated CG measurement system (±30 thousandths accuracy) that recommends ballast placement; deployed across hundreds of aircraft to ensure safe RATO launches, at an initial build cost of $2K.
- Supported development of a flagship Group 3 platform from concept to full-rate production in 4 months, integrating CRPA and BLOS data links, performing thermal analysis on the CRPA system, and supporting RATO system development.
- Integrated an EFI system into a Group 2 ISR platform, increasing range and performance; characterized and selected an optimal propeller to match the new powerplant.

Additive Manufacturing Manager — Florida Polytechnic University, Lakeland, FL — May 2022 — May 2024

- Utilized FDM and SLA additive manufacturing to fulfill project requests for professors, community members, and students.
- Collaborate with professors to support research through CAM software, generating G-Code for CNC Lathe and Mill operations.
- Responsible for managing a fleet of over 30 printers from brands like MakerBot, Stratasys, Prusa, Bambu, and Formlabs, efficiently overseeing the processing of more than 2000 prints annually, ensuring timely project completion, and achieving consistently exceptional outcomes.

Projects

Low-Cost Gimbaled RATO — Personal Project

- Developed the control loop for a Group 1/Group 2 gimbaled RATO solution integrated with ArduPlane, deriving the system's plant equation with support from a mechanical engineer; expanded allowable launch CG range to ±0.25", increasing payload/configuration flexibility.
- Designed a custom PCB and wrote C++ firmware for an ESP32-S3 controller to reduce cost, interfacing a current sensor, position sensor, servos, and MAVLink connection.
- Helped specify the electromechanical components required for the control loop, ensuring specs were mirrored in the control software for accurate tuning and launching.